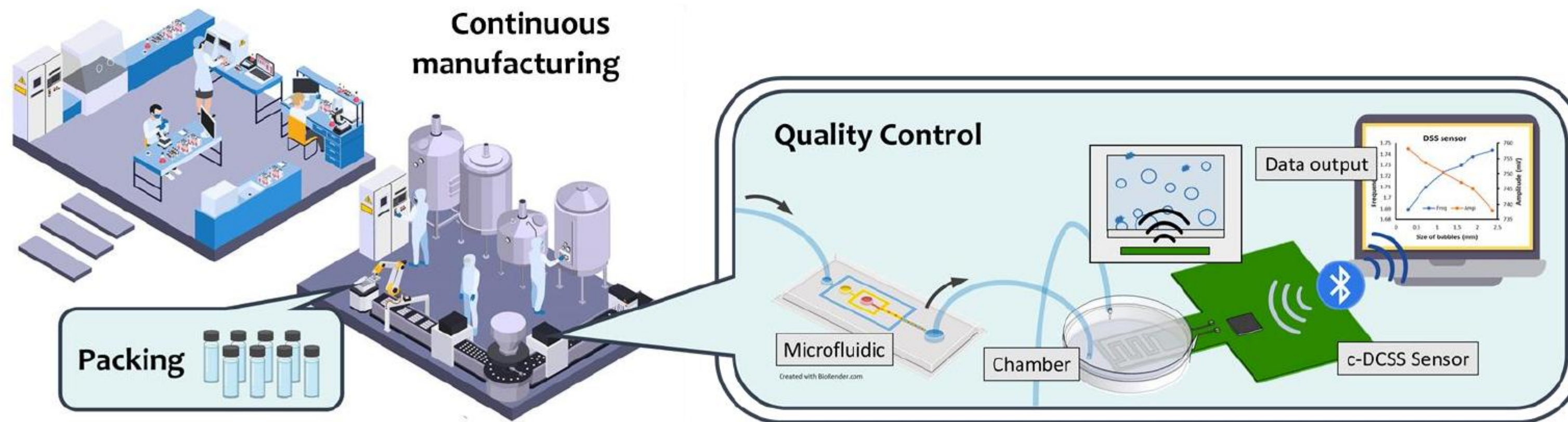


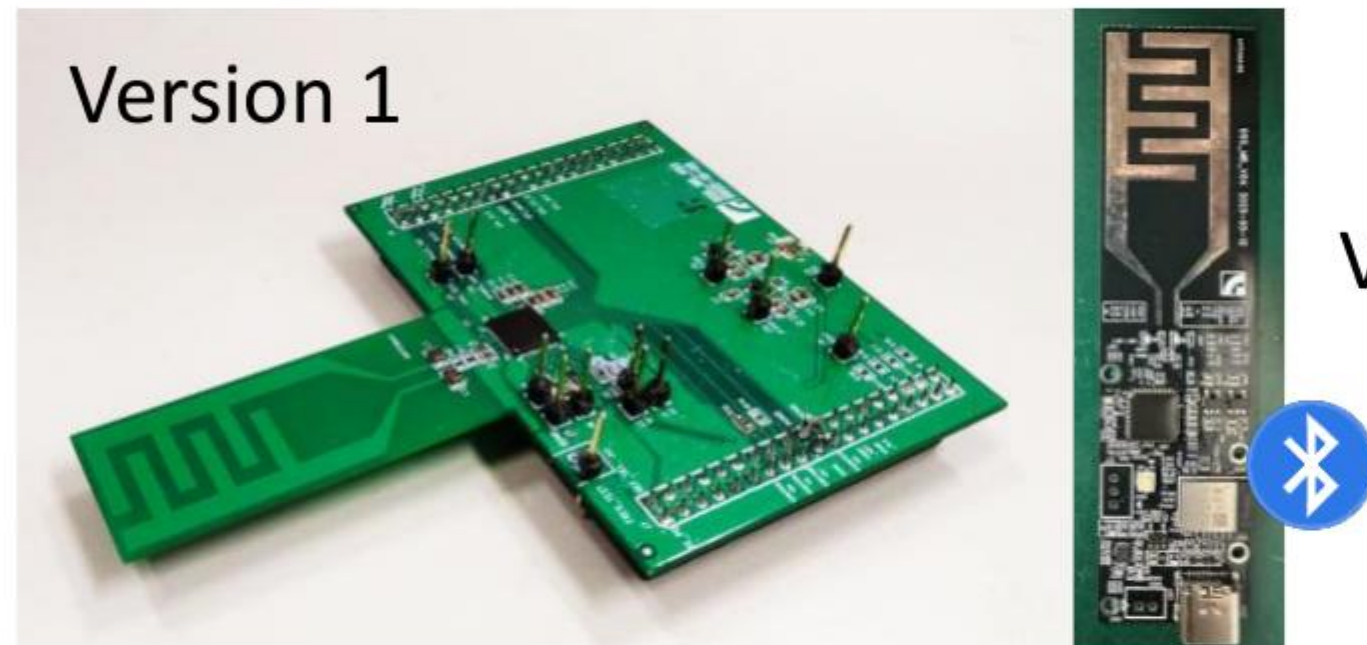
複數介電系數感測系統技術應用

Applications of Complex Dielectric Constant Sensor System (c-DCSS) Technology



Complex Dielectric Constant Sensor System

1. c-DCSS chip



3. What we measure & demonstrate

Basic measurements

- a. Ion Concentration
- b. Alcohol Concentration
- c. Sweetness (Brix)

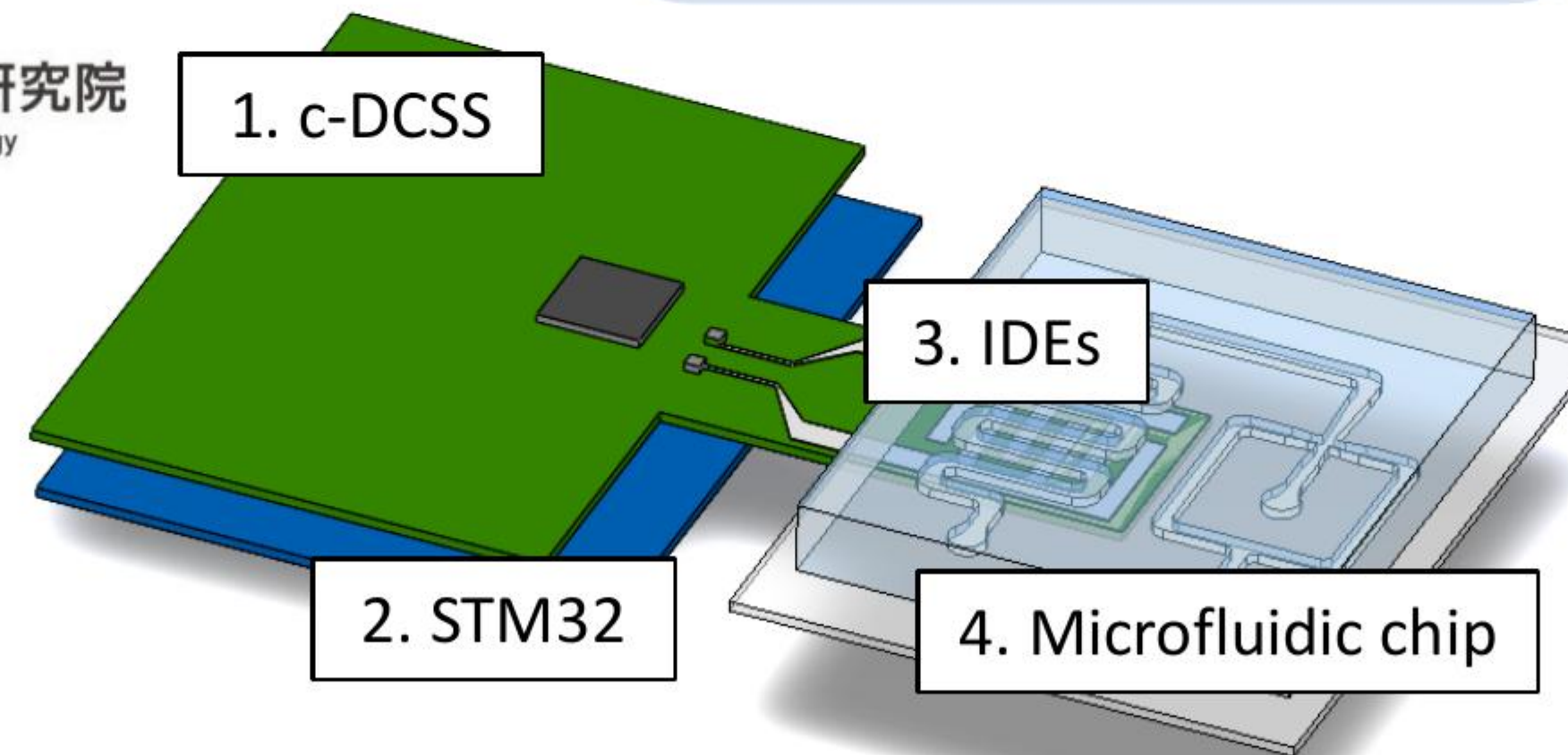
Advanced applications

- a. Real-Time Droplet Detection
- b. Packaged Sample Measurement

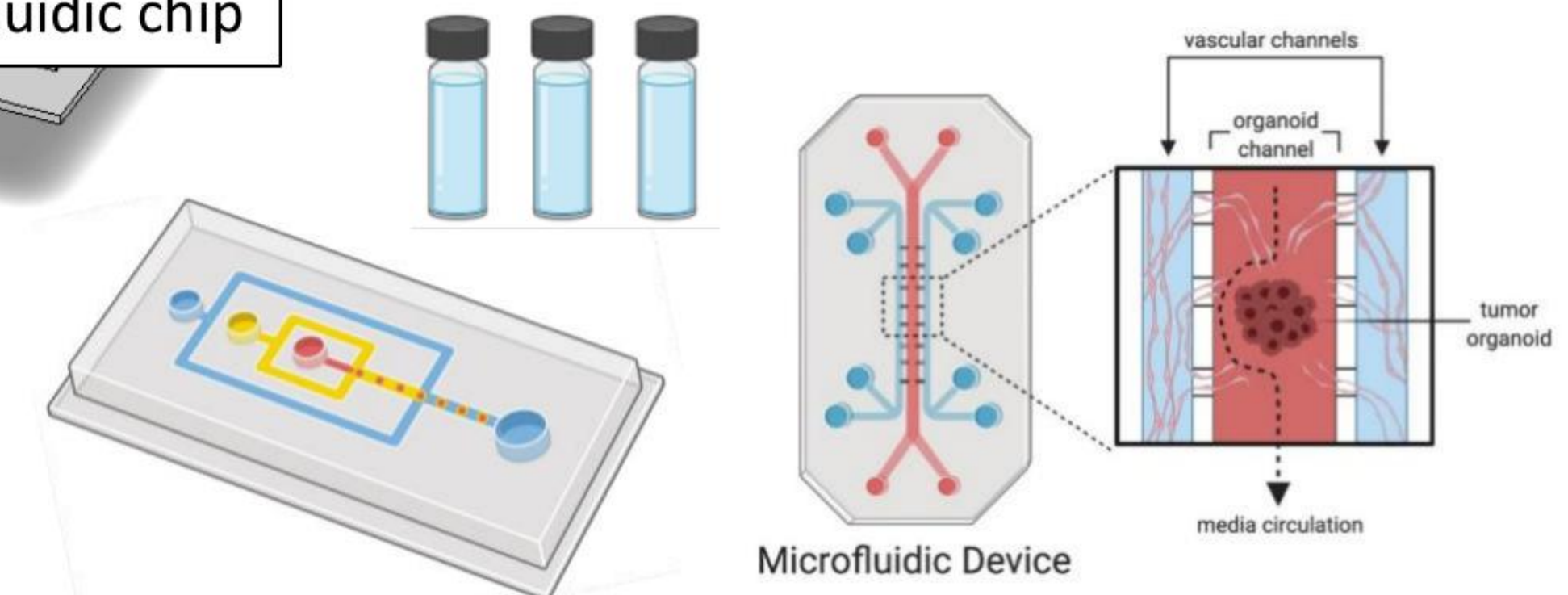
2. Firmware (MCU stm32)



- NUCLEO-L452RE
- WB5MMG (BT)

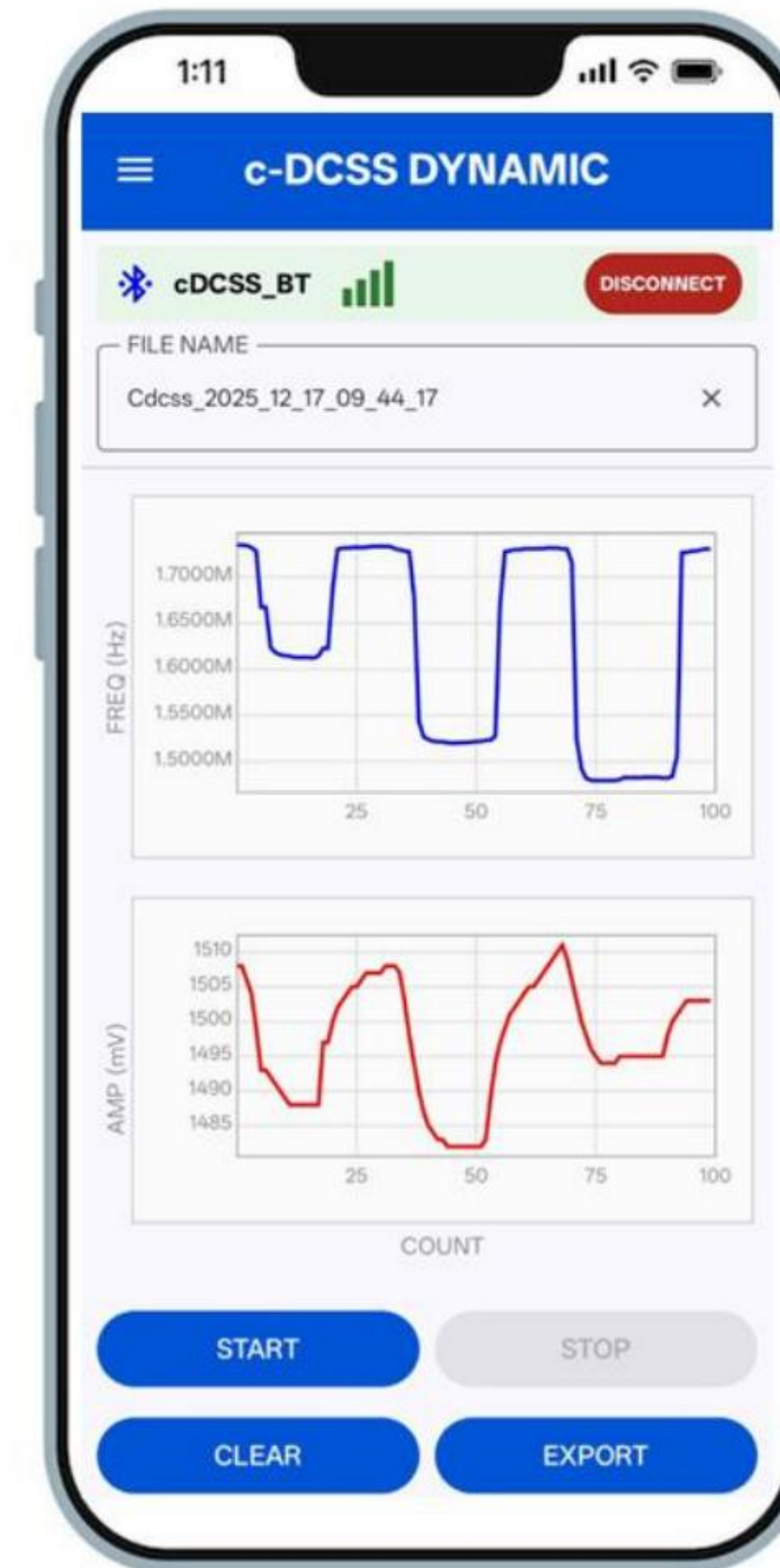
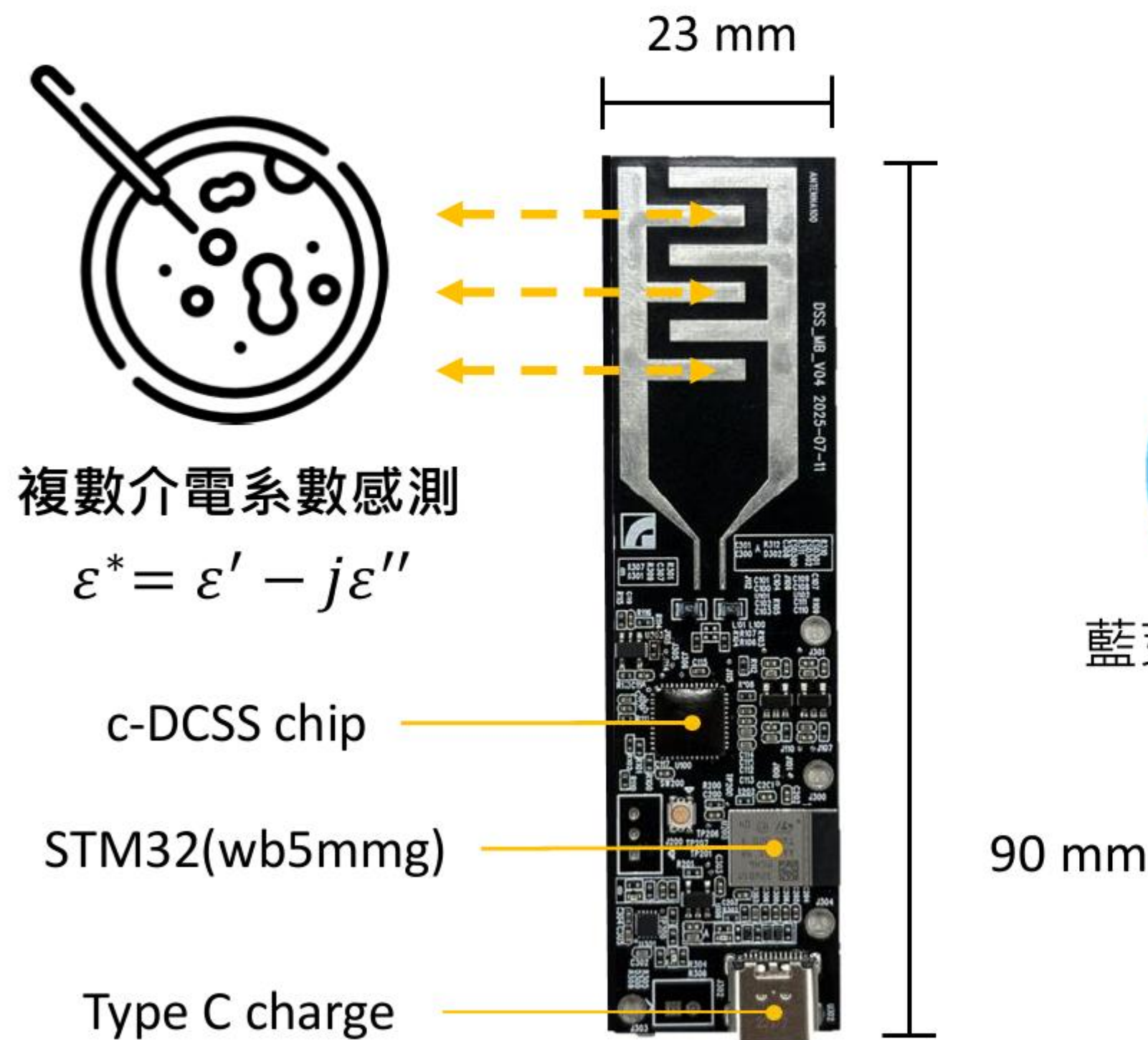


4. Process analytical technology



Complex Dielectric Constant Sensor System

✓ Non-Contact / Nondestructive testing ✓ Real-time response ✓ Low power consumption



動態即時顯示

介電系數實部顯示
 ϵ' ($\rightarrow$ Freq.)

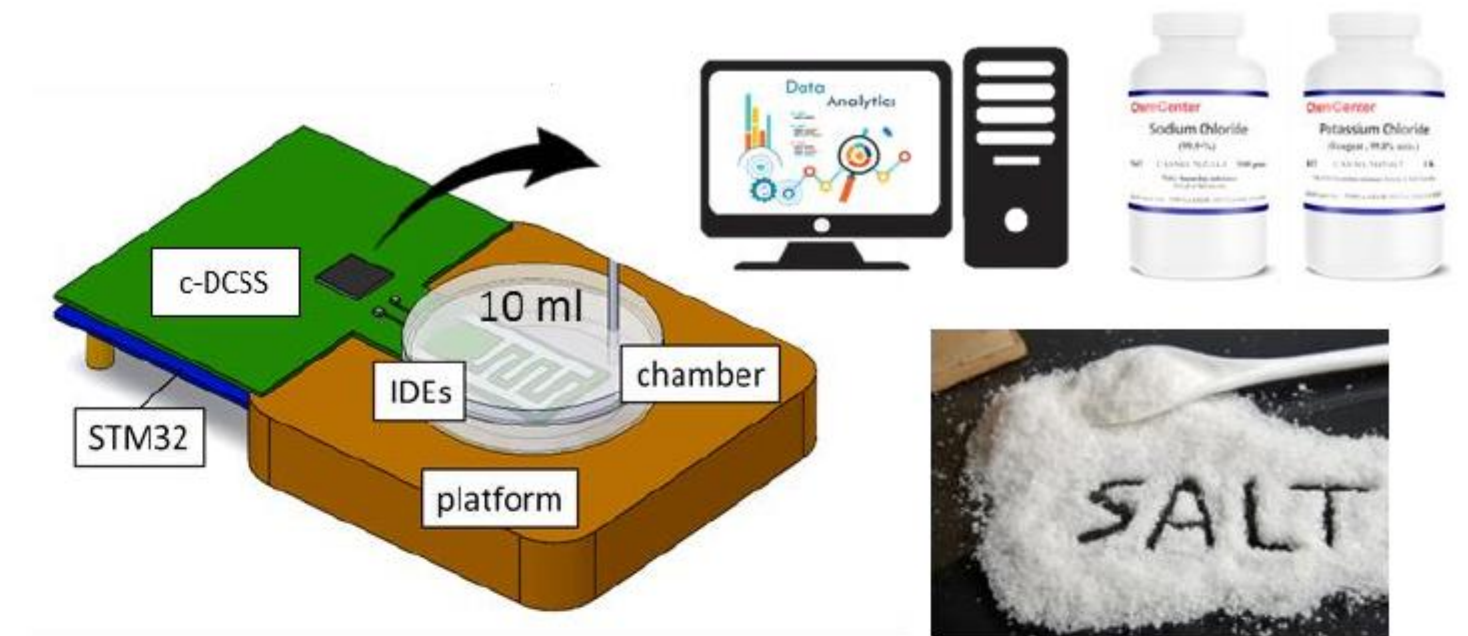
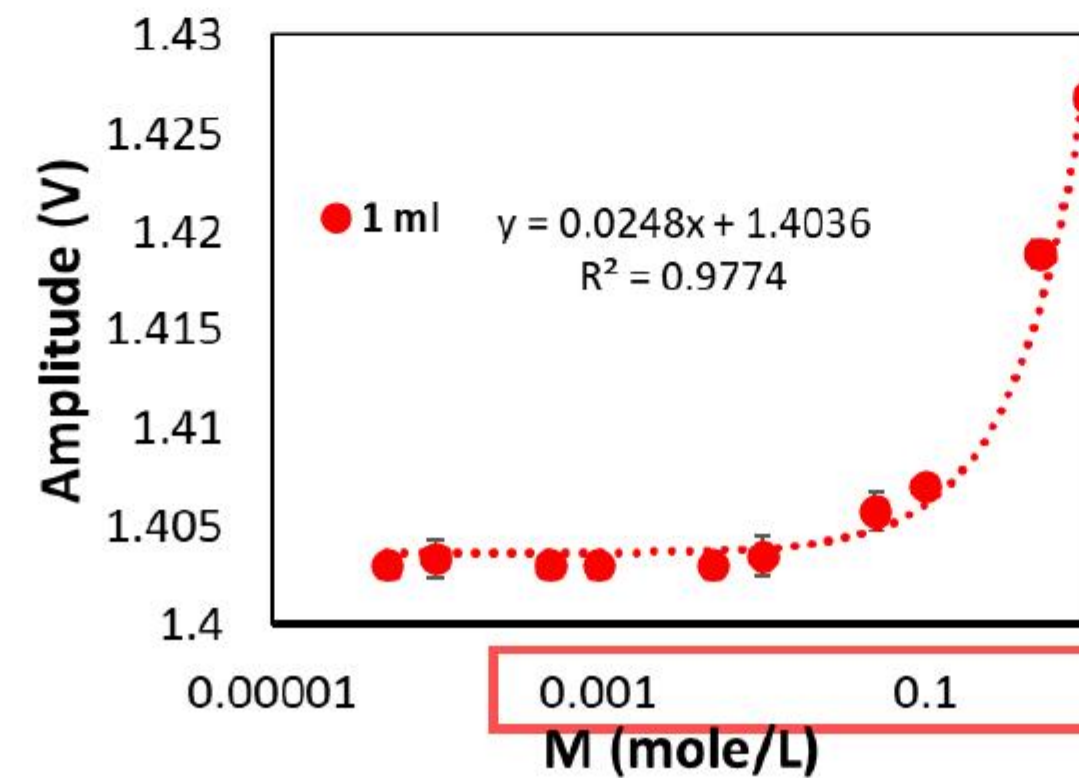
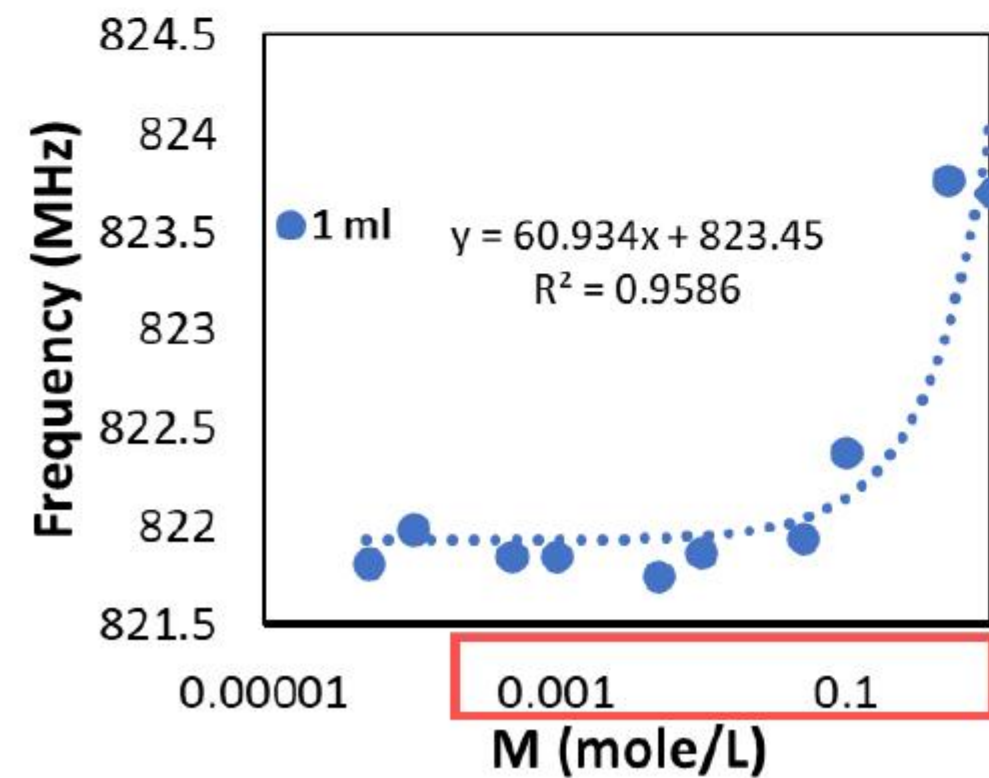
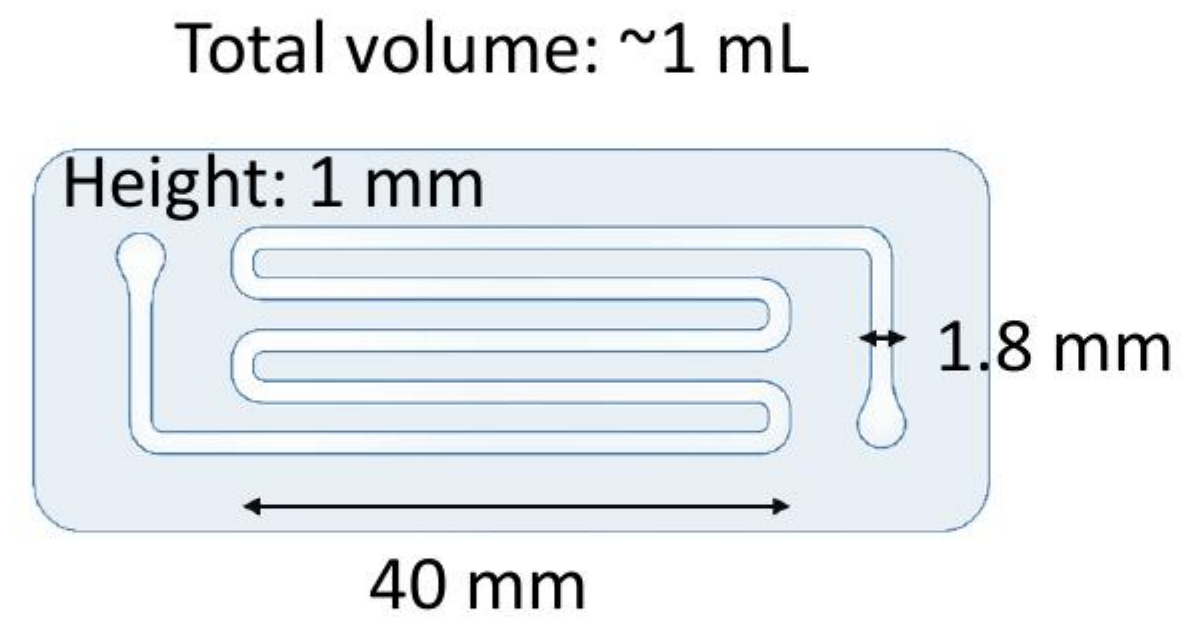
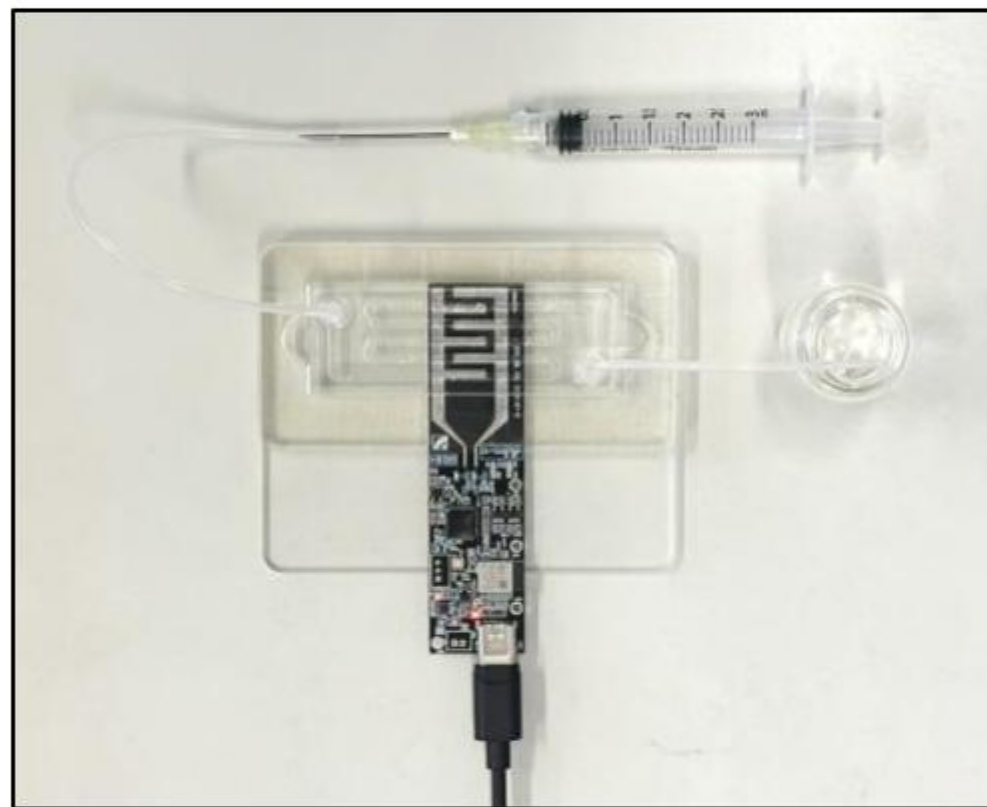
介電系數虛部顯示
 ϵ'' ($\rightarrow$ Amp.)

- ϵ' (real part): the alignment of dipoles
- $j\epsilon''$ (imaginary part): material loss
- The response frequency of c-DCSS is designed at 1 GHz.

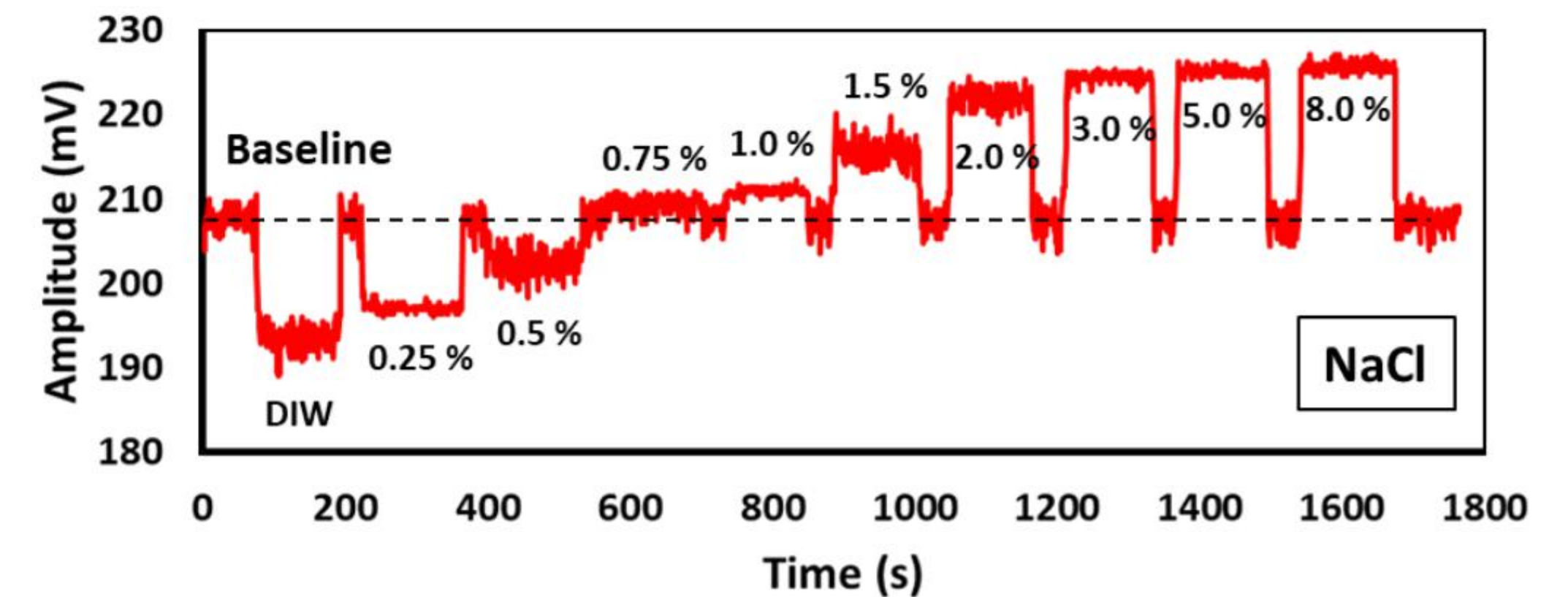
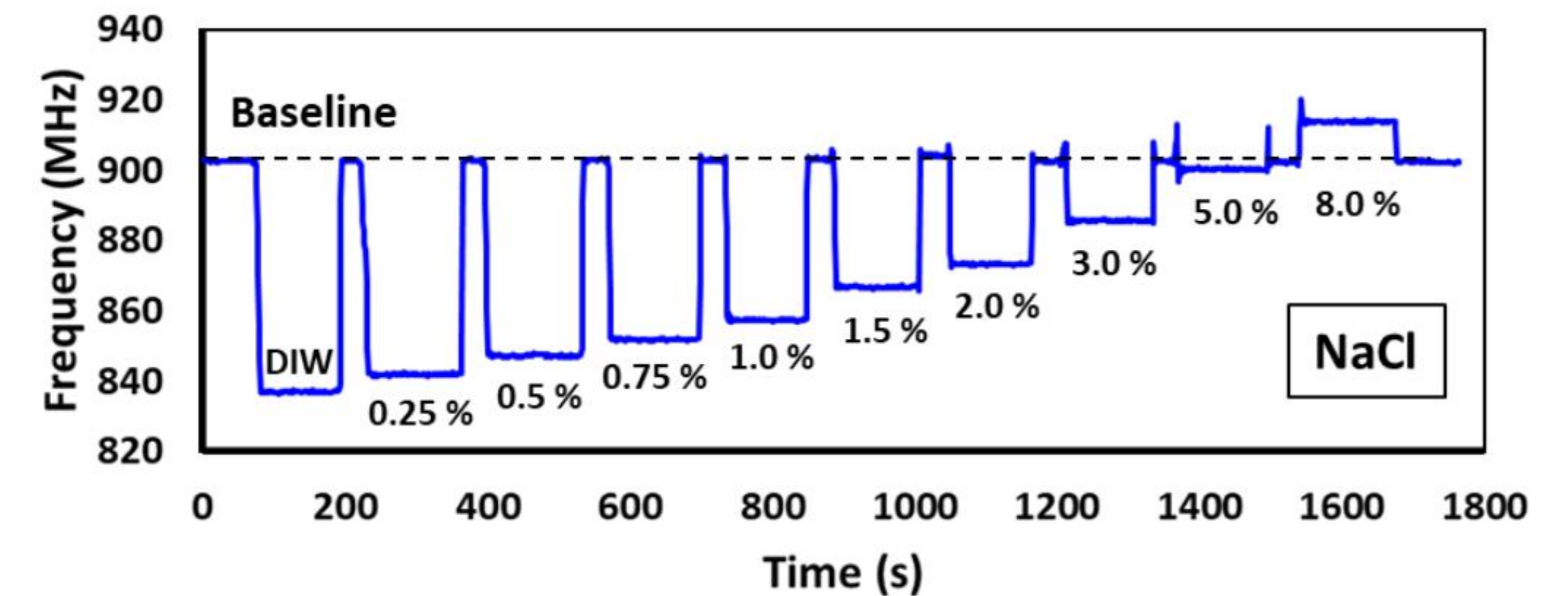
Results of c-DCSS measurements

Ion Concentration Measurement

- Dynamic range of ion concentration: 0.001~1 M

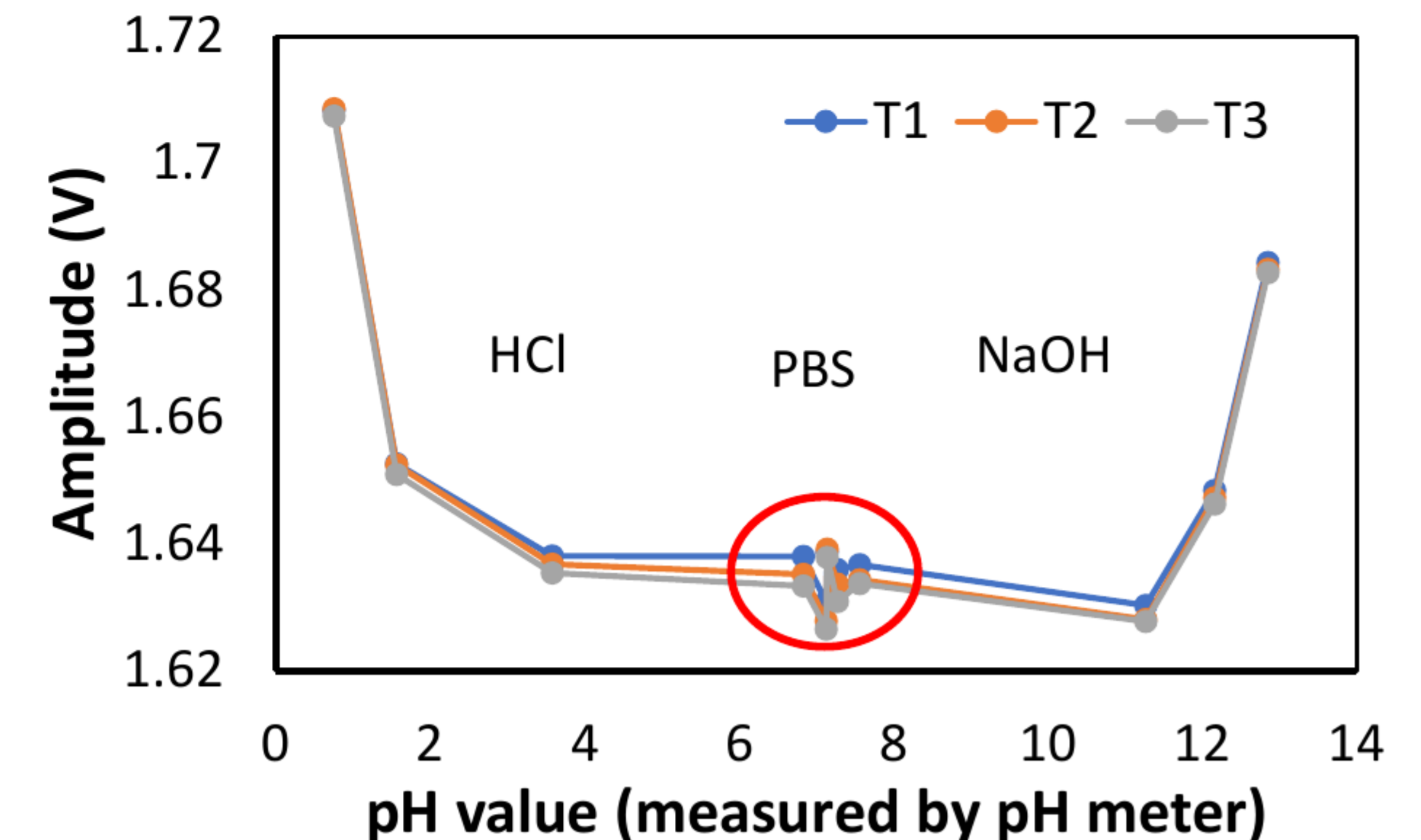
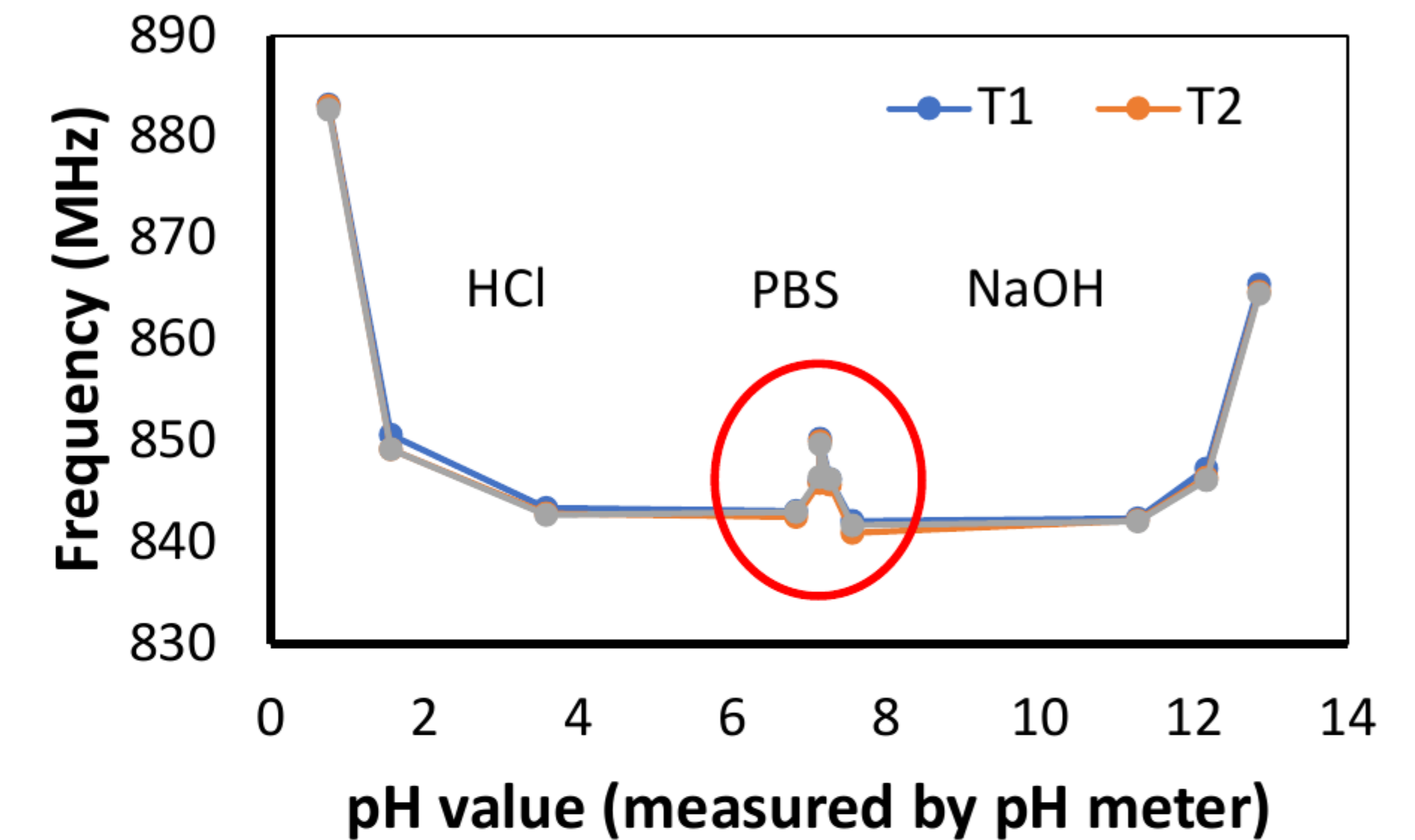
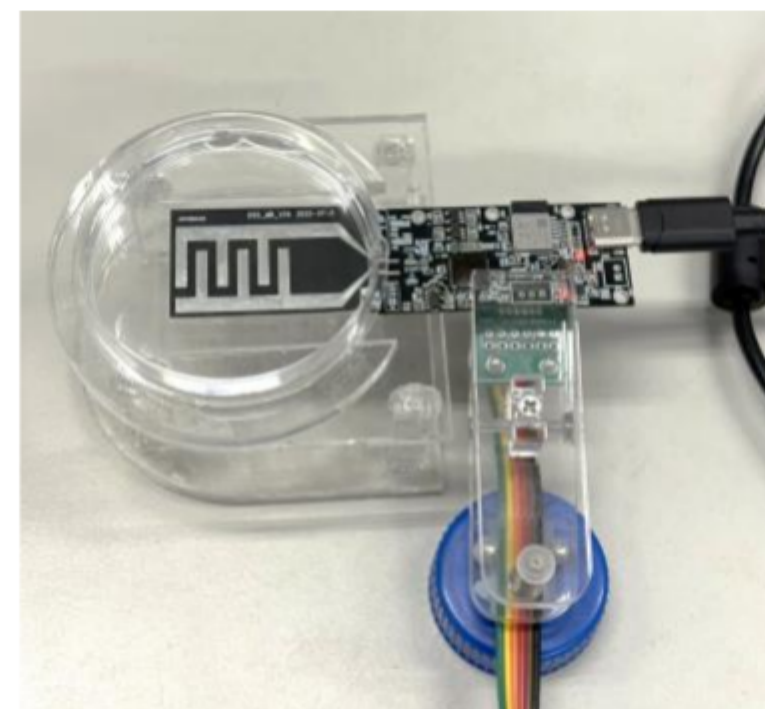


Dynamic measurement of ion concentration (KCl)



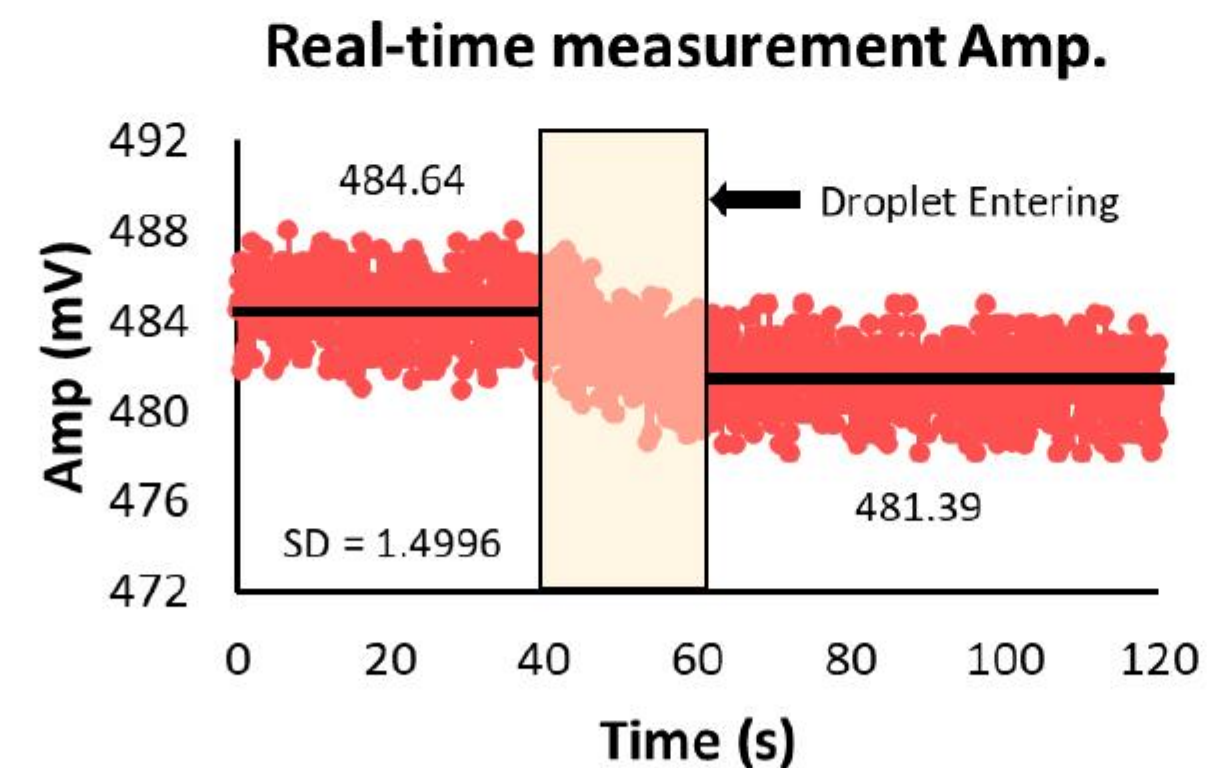
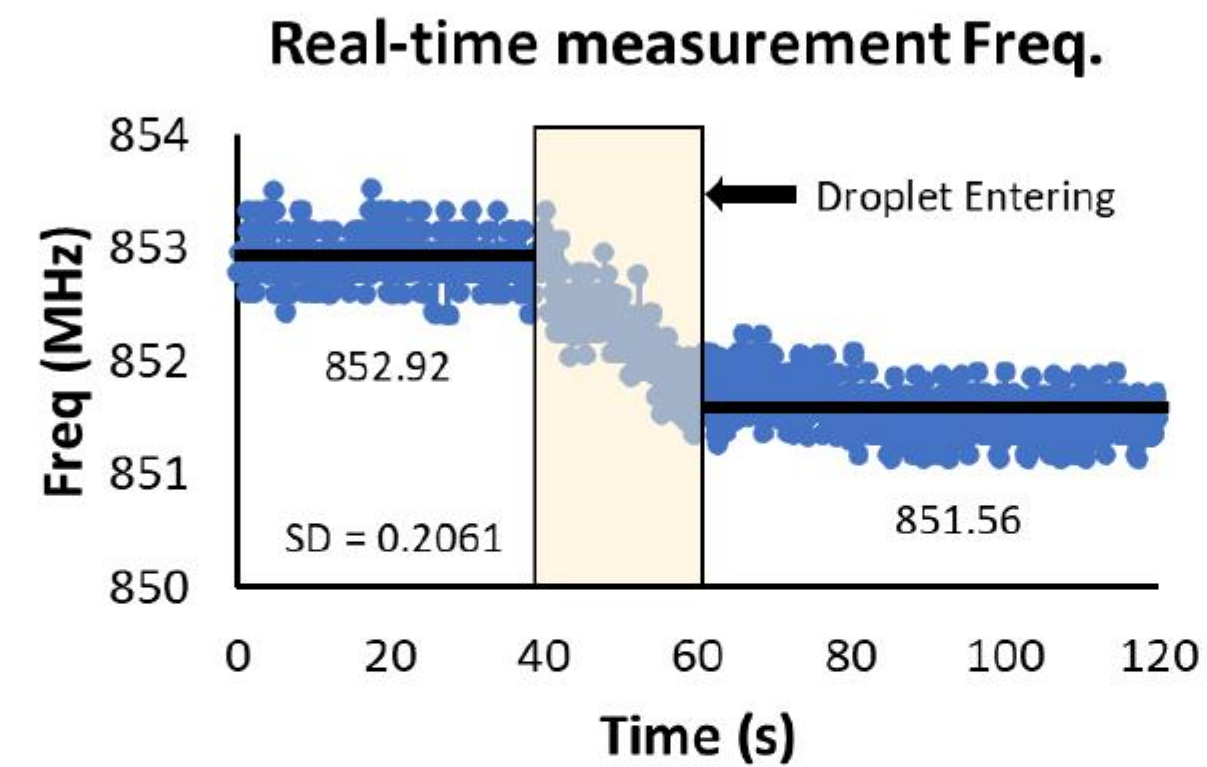
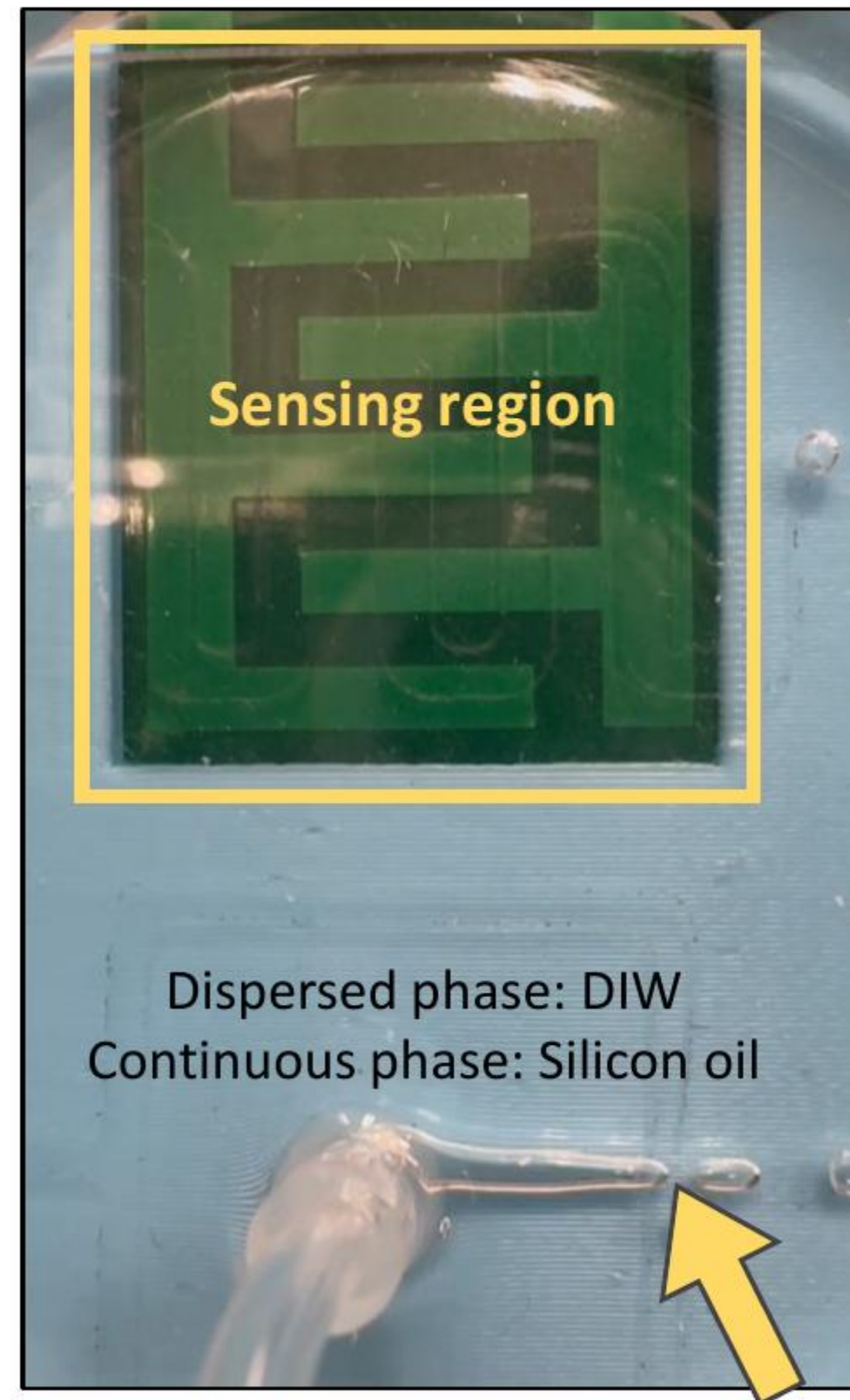
pH value measurement

- Mix 10 mL of 1x PBS with 2 mL of HCl/NaOH at different ratios.
 - 1x PBS 10 ml + 1 M HCl / NaOH 2 ml
 - 1x PBS 10 ml + 0.2 M HCl / NaOH 2 ml
 - 1x PBS 10 ml + 0.04 M HCl / NaOH 2 ml
 - 1x PBS 10 ml + 0.008 M HCl / NaOH 2 ml
 - 1x PBS 10 ml + 0.0016 M HCl / NaOH 2 ml
- pH value vs c-DCSS frequency (ϵ') and amplitude (ϵ'') readings
- c-DCSS can capture changes around pH 7.

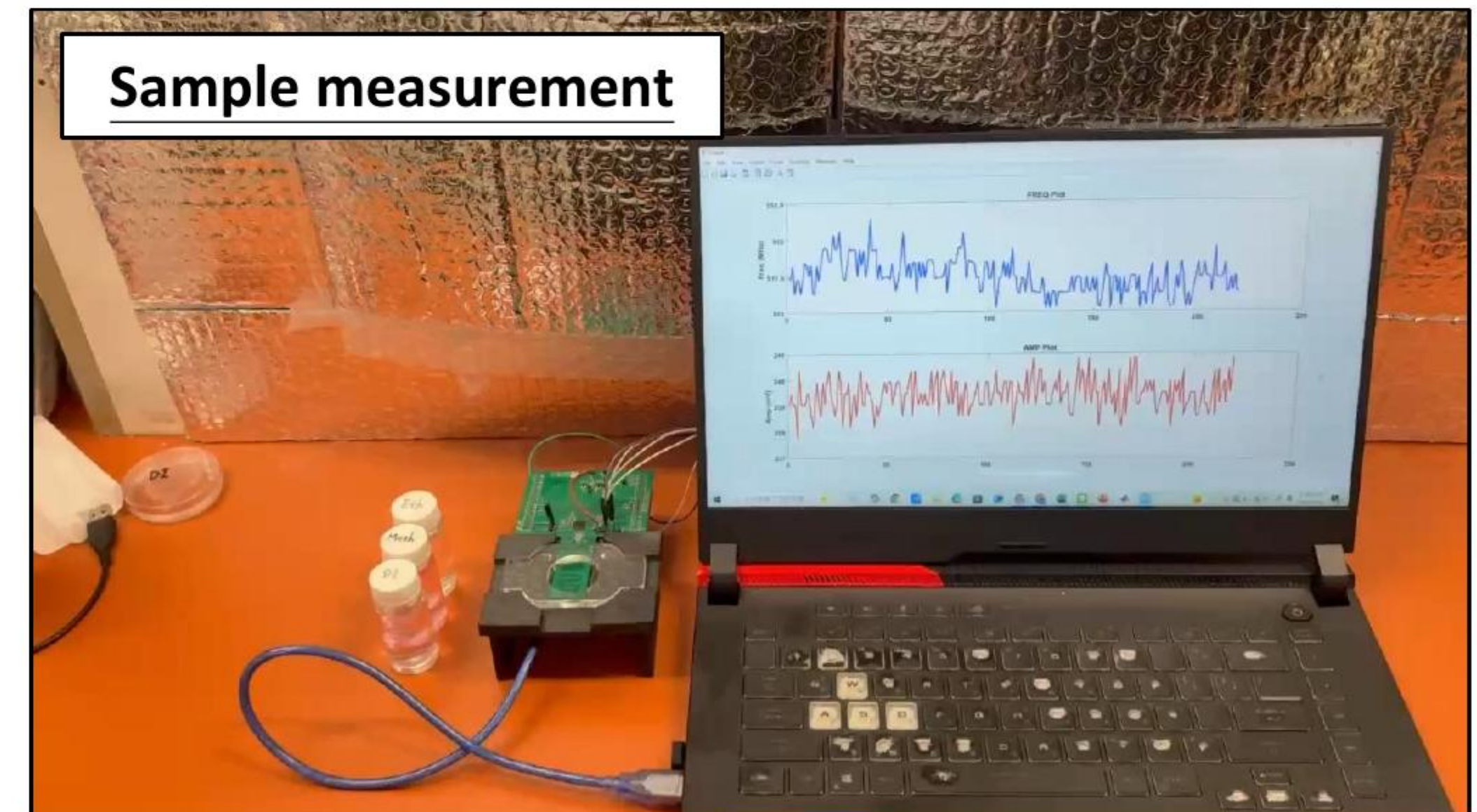
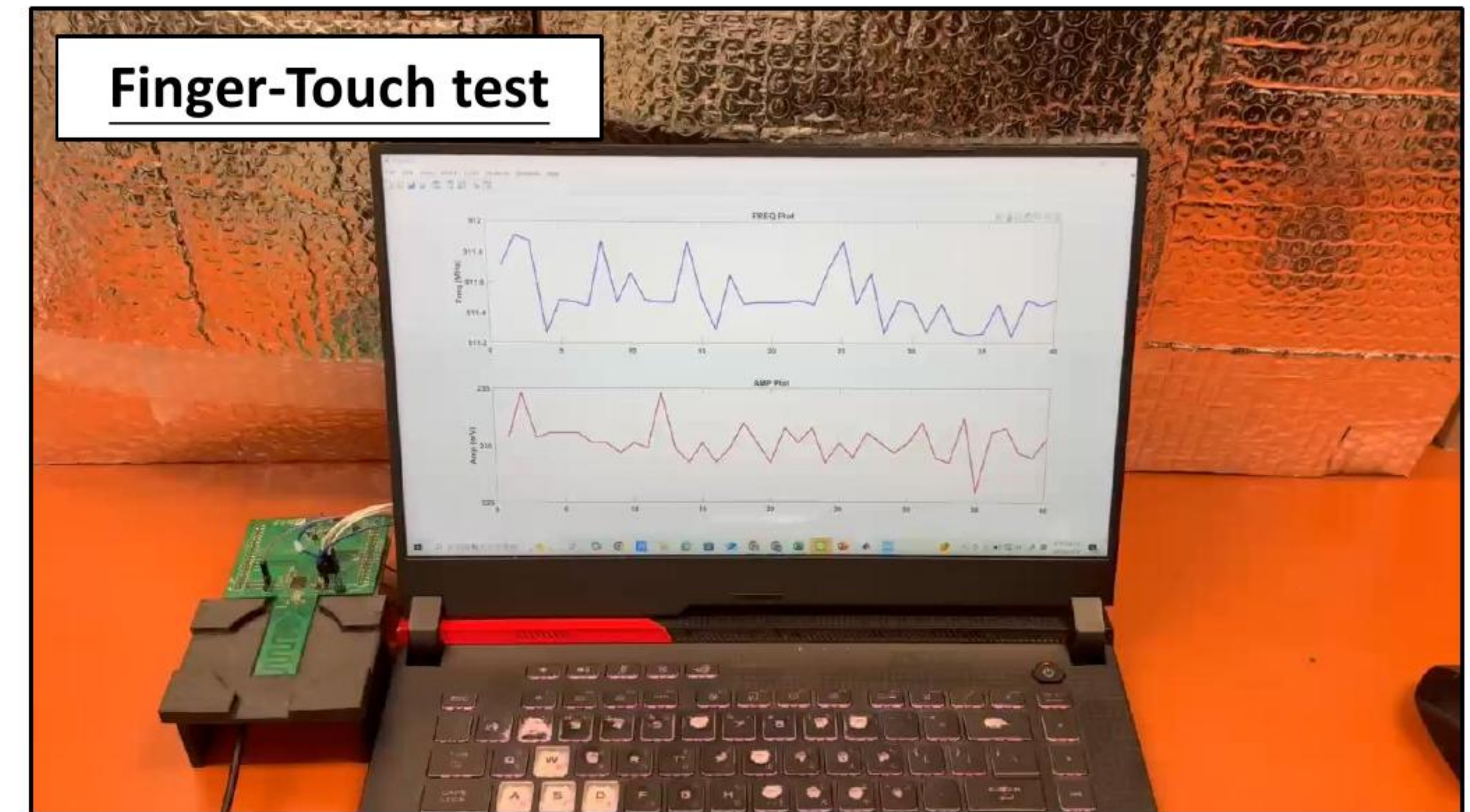


Demo Video of c-DCSS

Real-Time Droplet Detection

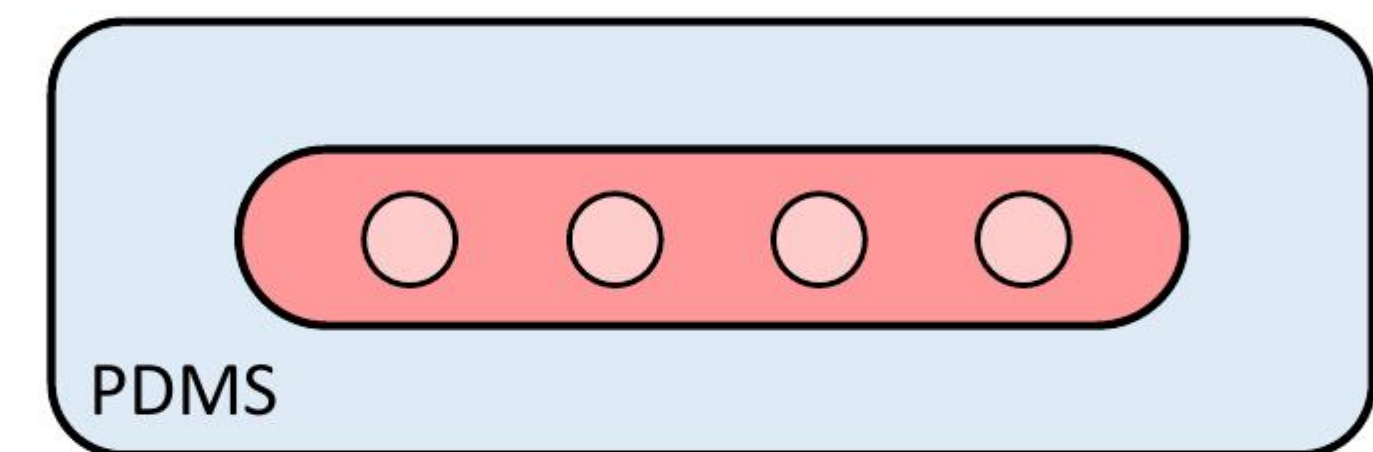
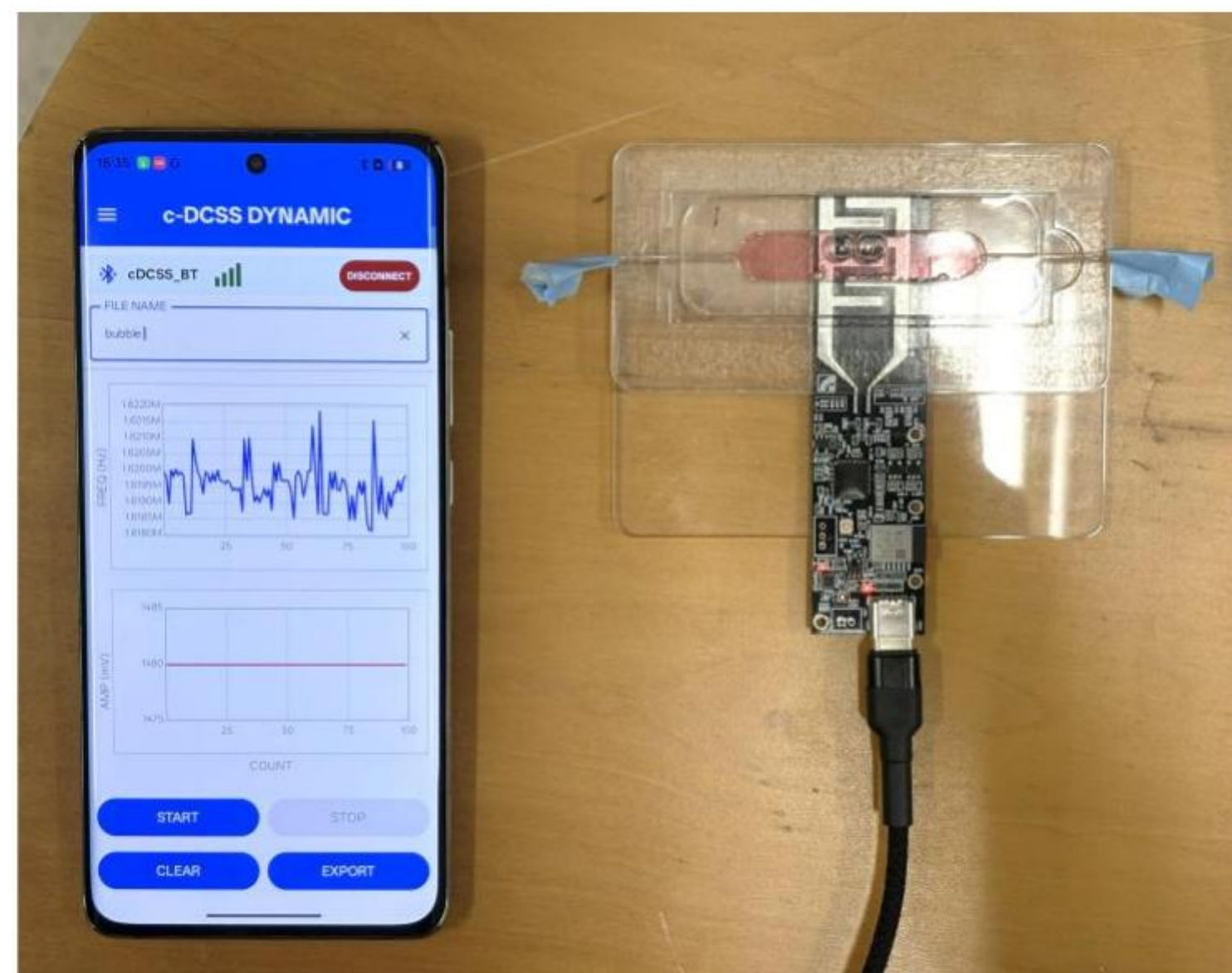


- (1) Initial stage (0~40 s): **Freq. = 851.56 MHz; Amp. = 484.64 mV**
- (2) Droplet entering stage (40~60 s): Freq. = -0.068 MHz/s; Amp. = -0.1625 mV/s
- (3) Stable production stage (60~120 s): **Freq. = 852.92 MHz; Amp. = 481.39 mV**

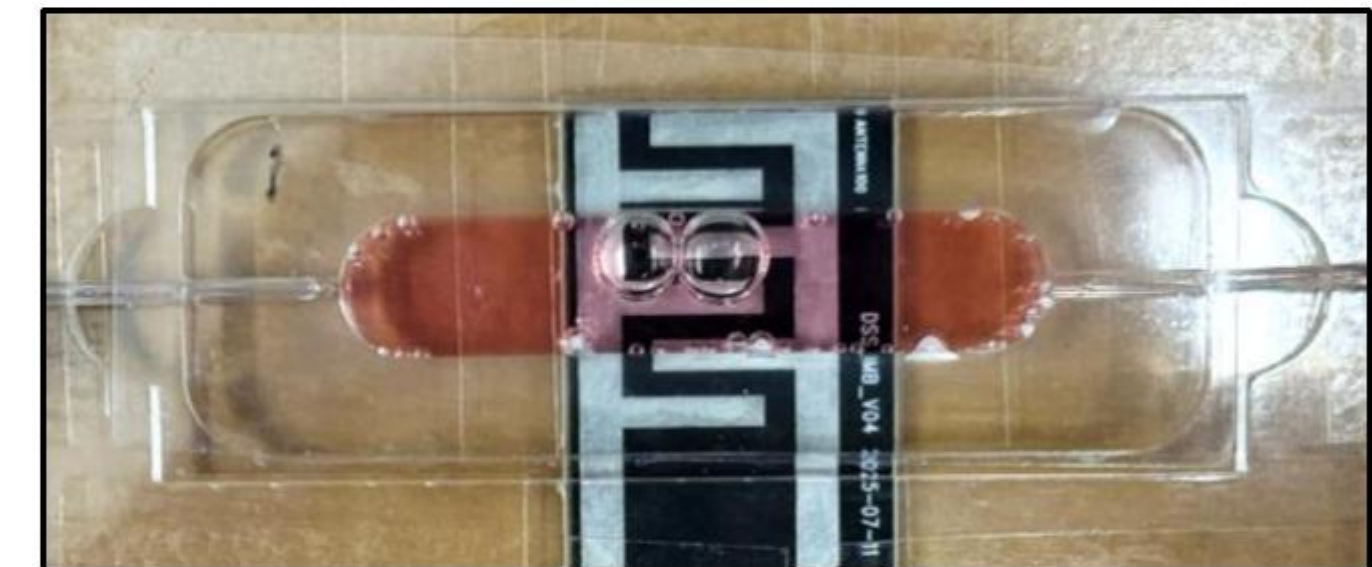


Cell culture experiment

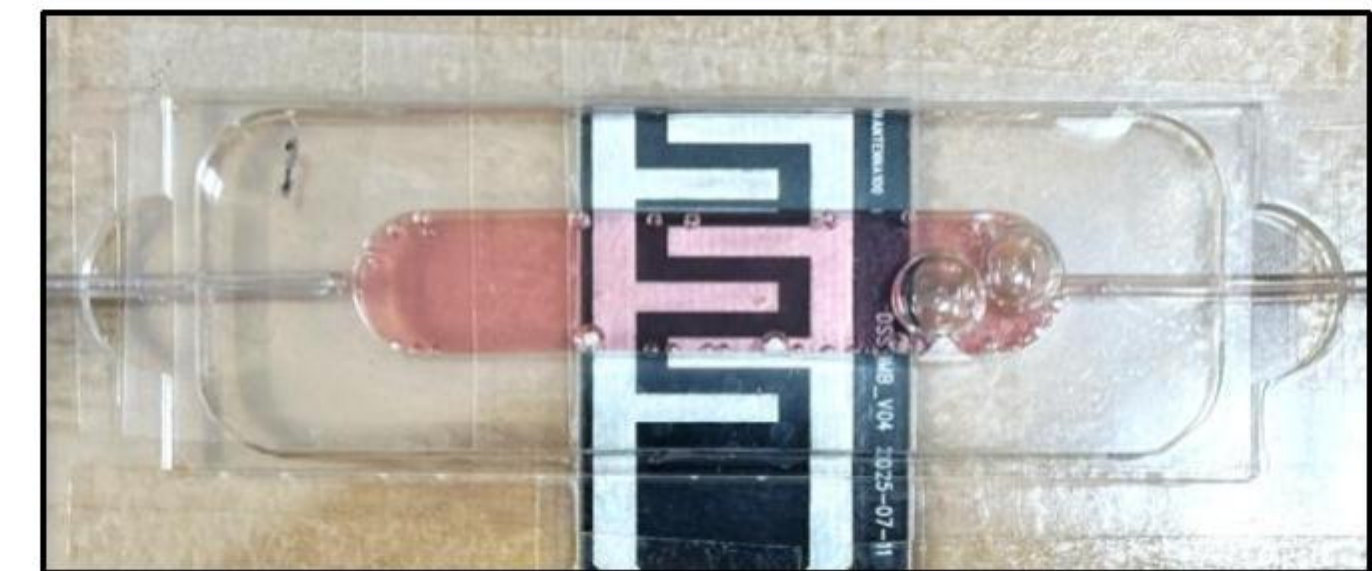
- Salivary gland cells
- Culture medium: DMEM
- Incubator: 37°C



Day 2



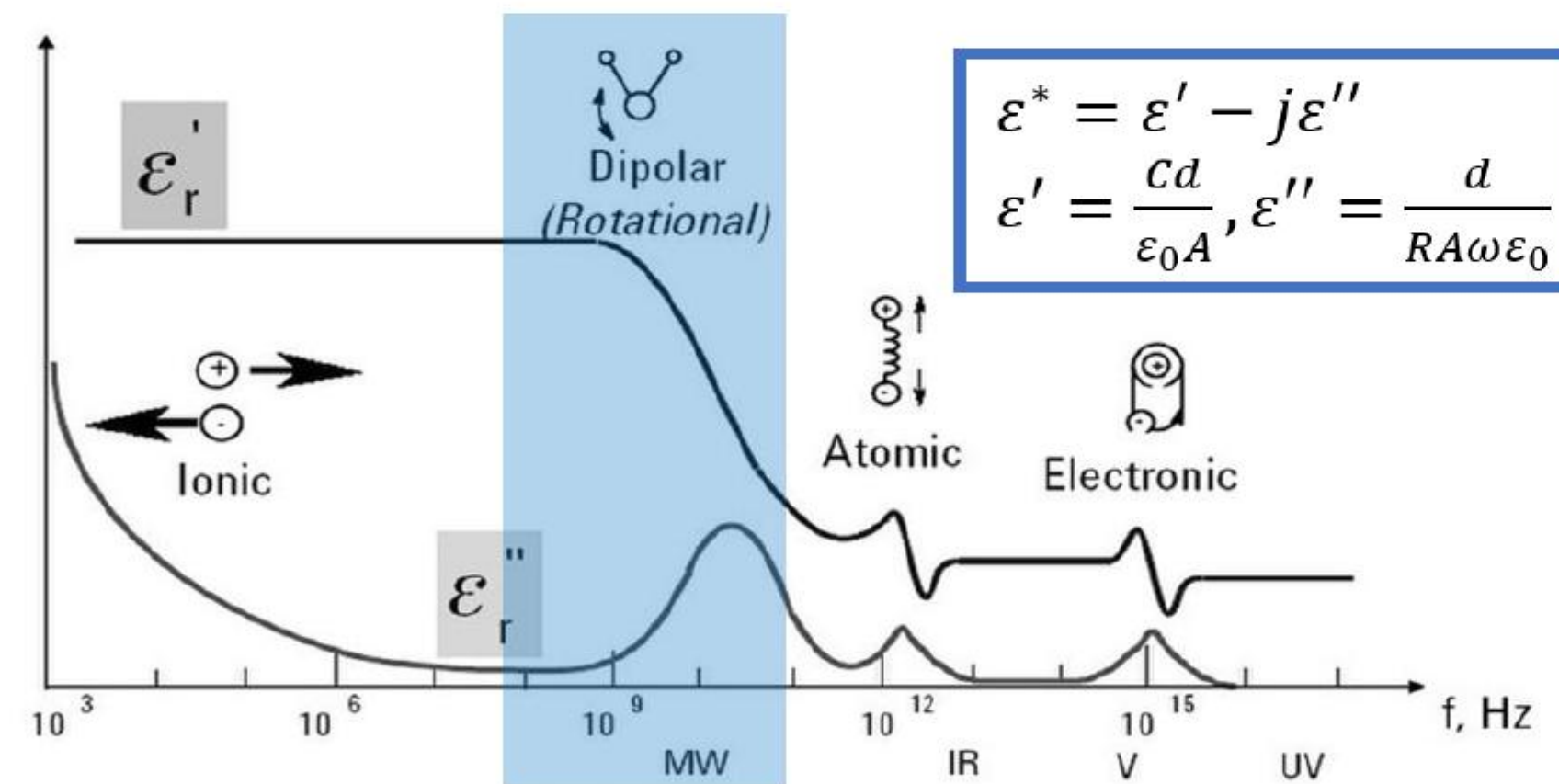
Day 3



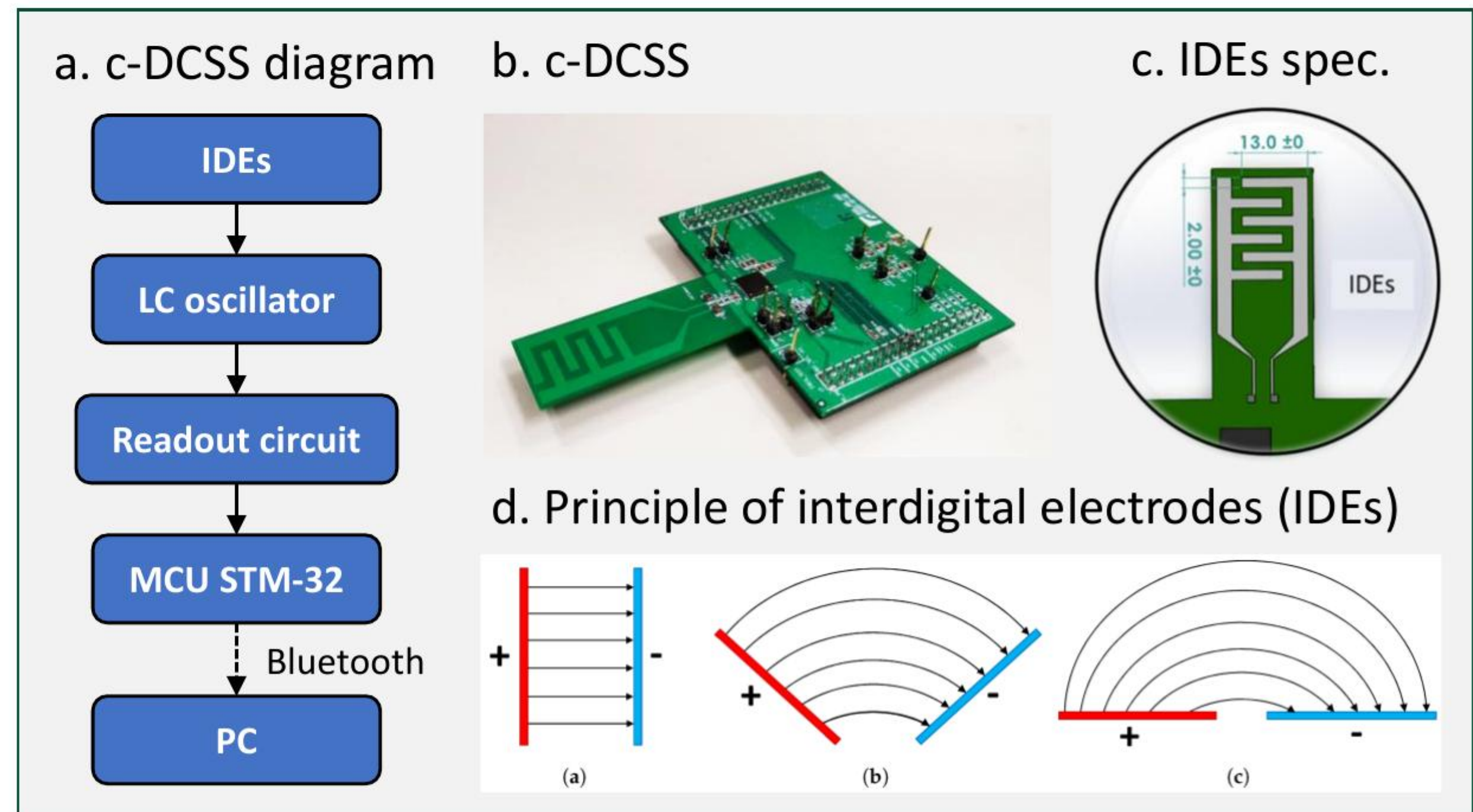
Complex Dielectric Constant Sensor System

✓ Non-Contact / Nondestructive testing ✓ Real-time response ✓ Low power consumption

- **Complex Dielectric Constant Sensor System (c-DCSS)** works by using microwave interactions with materials to measure their dielectric constant ($\epsilon^* = \epsilon' - j\epsilon''$)
- Sensing antenna is based on interdigital electrodes (IDEs) and inductors → LC oscillator (1 GHz)



- ϵ' (real part): the alignment of dipoles
- $j\epsilon''$ (imaginary part): material loss
- The response frequency of c-DCSS is designed at 1 GHz, which is suitable for measuring liquids samples.

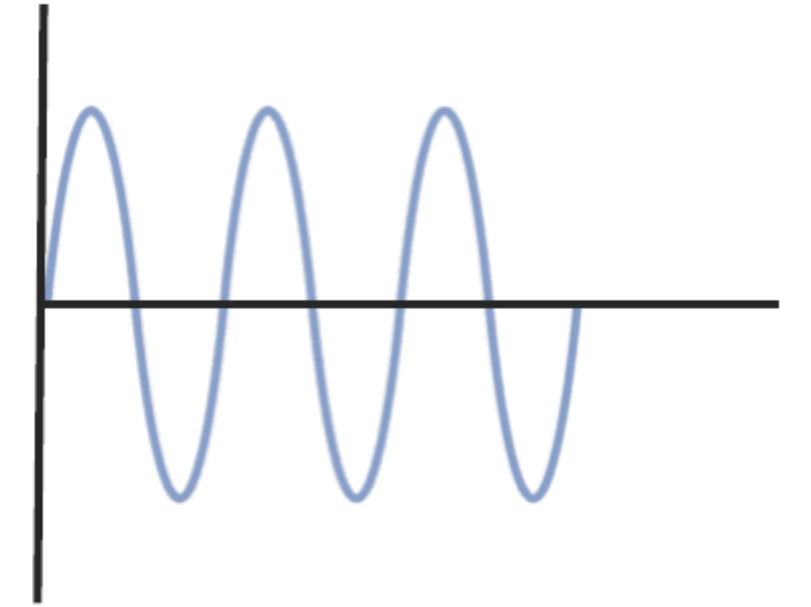


Complex Dielectric Constant Sensor System

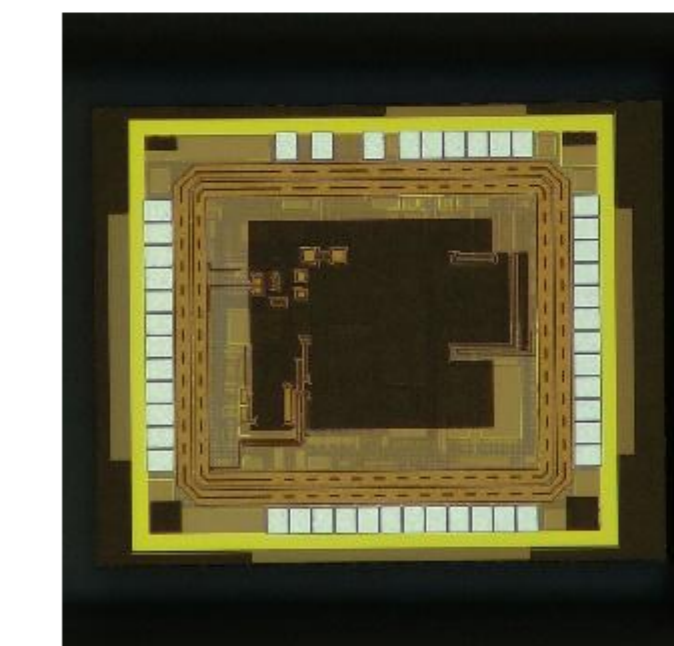
- IDEs (C) + inductor (L) $\rightarrow$ LC oscillator (1 GHz)
- The material under test (MUT) can be simplified as a parallel capacitor and inpendance in the circuit.

Initial state of c-DCSS

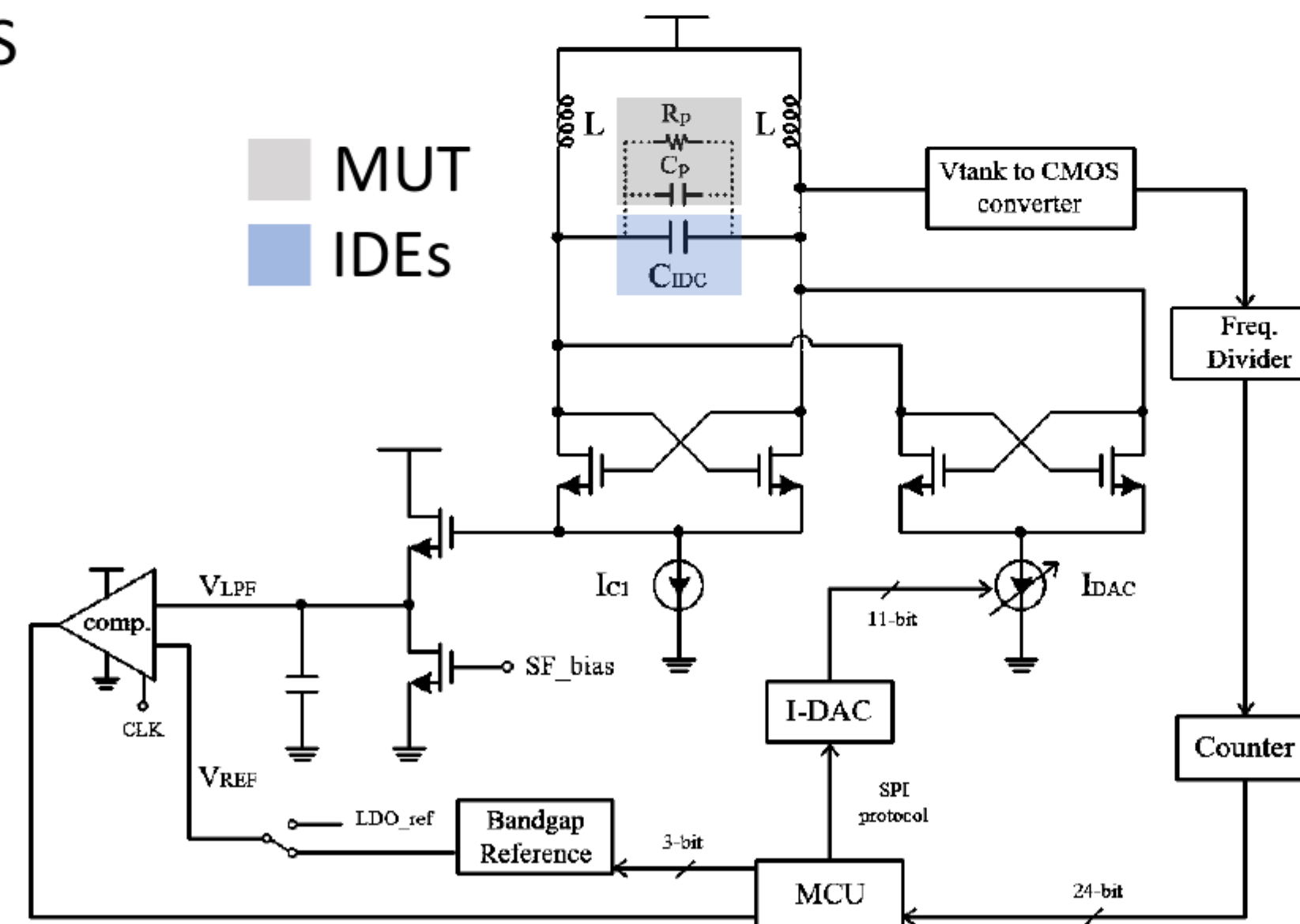
Signal produced by LC circuit



Schematic of c-DCSS



▲ TSMC 0.18 μm process

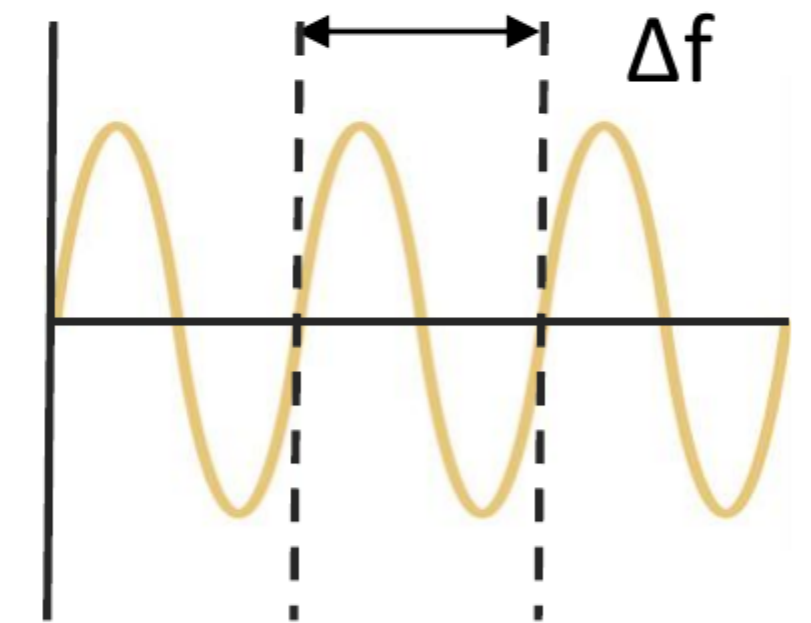


The c-DCSS consists of a CMOS dielectric sensor with frequency and amplitude readout circuits, fabricated by the TSMC 0.18 μm process.

Freq. readout (MUT)

Dielectric constant: ϵ'
 $\rightarrow$ frequency shift circuit

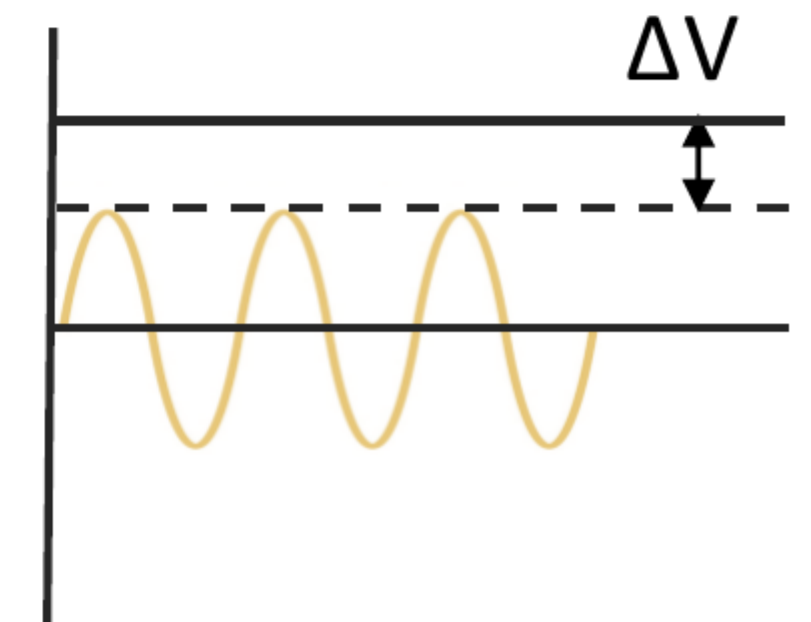
$$f = \frac{1}{2\pi\sqrt{LC}}$$



Amp. readout (MUT)

Dielectric constant: $j\epsilon''$
 $\rightarrow$ material loss circuit

$$V = \frac{4}{\pi} I_{tail} R_p$$



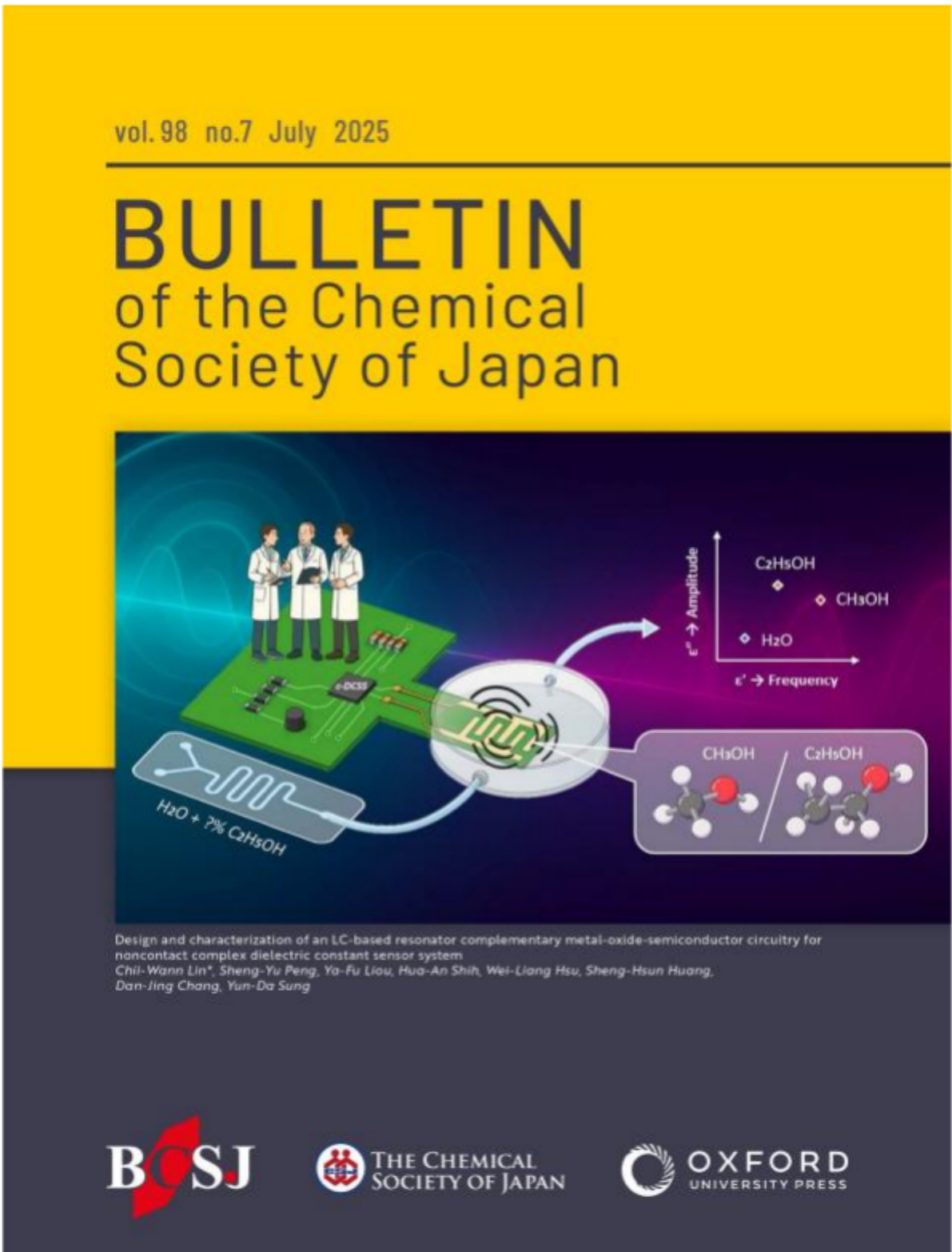
學術影響與技術佈局

學術期刊發表

- Lin CW, Peng SY, Liou YF, Shih HA, Hsu WL, Huang SH, Chang DJ, Sung YD. Design and characterization of an LC-based resonator complementary metal-oxide-semiconductor circuitry for noncontact complex dielectric constant sensor system. *Bull Chem Soc Jpn.* 2025;98(7):uoaf064. <https://doi.org/10.1093/bulcsj/uoaf064>
- Huang YJ, Wu HC, Chen PS, Shen HT, Peng SY, Lin CW. A non-invasive material sensing system and its integrated interface circuits. In: *Proc IEEE Int Symp Circuits Syst (ISCAS)*. Baltimore, MD, USA; 2017. <https://doi.org/10.1109/ISCAS.2017.8051008>
- Liou YF, Huang SH, Wu KD, Peng SY, Lin CW. (2024, November 17-20). Real-Time Measurement Micro-Bubbles Production in Microfluidics via Non-Contact Complex Dielectric Constant Sensor System [Oral presentation]. 2024 Asian Conference on Chemical Sensors, Kitakyushu, Japan.
- Huang SH, Lin XX, Liou YF, Lin CW. (2024, September 1-4). Design and Verification of An Automated Non-contact Di-electric Sensor System [Oral presentation] 2024 EUROSENSORS XXXVI, Debrecen, Hungary

專利佈局

序號	既有智財名稱	申請人	發明人	國別	專利號	案件狀態
1	介電常數量測電路及介電常數量測方法	國立臺灣大學、國立臺灣科技大學	林啟萬、彭盛裕、陳柏伸、林志鴻	TW	I495881	獲證
2	DELECTRIC CONSTANT MEASUREMENT CIRCUIT AND DELECTRIC CONSTANT MEASUREMENT METHOD	National Taiwan University, National Taiwan University of Science and Technology	Chii-Wann Lin, Sheng-Yu Peng, Bo-Shen Chen, Jhih-Hong Lin	US	US9423438 B2	獲證



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